

Finally, the watch is removed, and the Inotify instance is closed using the `inotify_rm_watch()` and `close()` function calls respectively.

Watching sub-directories

Unfortunately, there is no direct method to specify that you want all sub-directories under a particular directory to also be monitored. Hence, to watch sub-directories under a specified directory, you have to traverse each sub-directory and add a watch for each of them. Listing 2 does the above, traversing only the sub-directories immediately under the specified directory, using functions from `dirent.h`, and adds a watch for each. Subsequently, instead of printing events directly to `stdout`, it logs it into a file, `inotify_logger.log`.

Listing 2:

```
/* Using Inotify to monitor the sub-dirs under the specified dir*/

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <errno.h>
#include <sys/types.h>
#include <sys/inotify.h>
#include <dirent.h>
#include <limits.h>

#define MAX_LEN 1024 /*Path length for a directory*/
#define MAX_EVENTS 1024 /*Max. number of events to process at one go*/
#define LEN_NAME 16 /*Assuming that the length of the filename won't exceed 16 bytes*/
#define EVENT_SIZE ( sizeof (struct inotify_event) ) /*size of one event*/
#define BUF_LEN ( MAX_EVENTS * ( EVENT_SIZE + LEN_NAME ))
/*buffer to store the data of events*/

/* Log file*/
FILE *fp_log;

/* Add inotify watches to directories immediately under root
 * in addition to itself */

void add_watches(int fd, char *root)
{
    int wd;
    char *abs_dir;
    struct dirent *entry;
    DIR *dp;

    dp = opendir(root);
    if (dp == NULL)
```

```
{
    perror("Error opening the starting directory");
    exit(0);
}

/* add watch to starting directory */
wd = inotify_add_watch(fd, root, IN_CREATE | IN_MODIFY | IN_DELETE);
if (wd == -1)
{
    fprintf(fp_log, "Couldn't add watch to %s\n", root);
}
else
{
    printf("Watching:: %s\n", root);
}

/* Add watches to the Level 1 sub-dirs*/
abs_dir = (char *)malloc(MAX_LEN);
while((entry = readdir(dp)))
{
    /* if its a directory, add a watch*/
    if (entry->d_type == DT_DIR)
    {
        strcpy(abs_dir, root);
        strcat(abs_dir, entry->d_name);

        wd = inotify_add_watch(fd, abs_dir, IN_CREATE | IN_MODIFY | IN_DELETE);
        if (wd == -1)
            printf("Couldn't add watch to the directory %s\n", abs_dir);
    }
    else
        printf("Watching:: %s\n", abs_dir);
}

closedir(dp);
free(abs_dir);
}

/* Main routine*/
int main( int argc, char **argv )
{
    int length, i = 0;
    int fd;
    char buffer[BUF_LEN], root[MAX_LEN];

    /*Check for supplied path to monitor*/
    switch(argc)
    {
        case 1: printf("No directory specified. Will monitor the entire filesystem...\n\n");
                strcpy(root, "/");
```